

Question	Marking Guidance	Mark	Comments
03.1	Max 5 from: 1. DNA helicase unwinds DNA/double helix OR DNA helicase separates strands OR DNA helicase breaks hydrogen bonds; 2. Both strands act as templates; 3. (Free DNA) nucleotides line up in complementary pairs/A-T and G-C; 4. DNA polymerase joins nucleotides (of new strand); 5. Forming phosphodiester bonds; 6. (Each new DNA molecule consists of) one old/original/template strand and one new strand;	5 max (5 x AO1)	1. Reject hydrolyses hydrogen bonds 1. Accept ‘unzip’ for unwind 2. Accept description of ‘template’, eg exposed bases on single (polynucleotide) strands 4. Reject forms hydrogen bonds/joins bases

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03.2	1. (Produced by) semi-conservative replication OR One old/original strand and one new strand; 2. (Contains) ^{14}N/light (strand) and ^{15}N/heavy (strand);	2 (2 x AO2)	

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03.3	One band level with band in sample 2 and one band higher, drawn onto the sample 3 tube;	1 (1 x AO2)	Allow lines of any thickness Ignore lines drawn on the other tubes Both lines needed for mark